

Designing Credible Evidence of Learning

AI-aware summative assessment options at Oxford

— Start with validity, not resistance

The aim is not to make an assessment universally “AI-proof”. The aim is to obtain credible, inclusive and proportionate evidence of the intended learning while communicating AI permissions clearly. This guide replaces absolute language such as “inherently AI-resistant” with a more defensible account of what different formats can evidence, what risks remain and which safeguards may be proportionate. The full interactive catalogue remains available in the Oxford AI Toolkit.

ASSESSMENT-DESIGN PRINCIPLE

Design assessments that produce verifiable judgement.

The central question is not simply “Did AI write this?” but whether the assessment provides credible evidence that the student can reach a reasoned conclusion, and explain, defend and apply it. VIVA is the practical framework for designing that evidence.

— The design standard

1 Valid

The task and conditions produce evidence of the knowledge, reasoning, judgement, performance or professional capability intended.

2 Inclusive

Access, disability, language, digital confidence, time, cost and reasonable adjustments are considered.

3 Visible

The student’s reasoning, decisions, verification or performance are visible through relevant evidence.

4 Accountable

Permissions, declaration, verification and responsibility for the final submission are explicit.

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— Choose evidence before format

WHAT MUST BE EVIDENCED?	USEFUL EVIDENCE SOURCES	COMMON VALIDITY RISK	DESIGN RESPONSE
Foundational knowledge	Controlled recall, explanation, application	Fluent output masks weak understanding	Use questions that require selection, explanation and transfer.
Reasoning and judgement	Justified decisions, trade-offs, critique, application	Generic analysis receives high marks	Reward evidence, assumptions, alternatives and disciplinary judgement.
Practical performance	Observed action, product, decision under conditions	Written account substitutes for performance	Observe the relevant capability; use standardised criteria.
Research process	Milestones, methods, data, source decisions, reflexivity	Process evidence becomes surveillance or paperwork	Collect only evidence that supports learning and validity.
Communication	Audience adaptation, live response, visual/oral choices	Polish overwhelms substance	Separate content, communication and response criteria.
Responsible AI use	Verification, critique, limitations, disclosure	Marks reward prompt fluency or tool access	Provide equal access and assess evaluation and accountability.

IMPORTANT LIMITATION

Controlled conditions, original data, process records, oral questioning and live performance can strengthen particular kinds of evidence. None automatically proves authorship, eliminates unauthorised assistance or removes the need for fair procedures and professional judgement. Where AI could materially obscure the capability being assessed, draw on more than one proportionate source of evidence, such as the submitted work together with staged drafts, explanation or dialogue. Triangulation strengthens the judgement; no single component proves authorship.

— Five reusable design patterns

Controlled independent performance

Use when unaided performance under specified conditions is itself important. Strengthen with relevant application, explanation and reliable approved conditions. Do not assume online controls eliminate access to other devices.

Contextual application and judgement

Use supplied evidence, disciplinary constraints, stakeholders and trade-offs. Specificity helps validity because it demands engagement, not because the material is assumed to be outside a model's training data.

Visible process and decisions

Use proposals, milestones, notebooks, drafts, code history or reflective notes when development and decision-making are part of the learning. Keep requirements proportionate and avoid treating logs as proof.

Explanation and defence

Use written rationale, live response or structured questioning to probe understanding. Pre-specify assessed oral components, standardise criteria and provide reasonable adjustments.

AI-integrated evaluation

Require students to compare, verify, critique or improve AI output. Provide equal access, define tool contingencies and reward disciplinary judgement rather than surface fluency.

— Assessment-format guide

In-person timed examination

Can evidence: Independent knowledge, reasoning or calculation under controlled conditions.

Design for visibility: Use unseen application, workings, explanation or interpretation. State AI is not permitted if that is the rule.

Caution: Controlled conditions strengthen independent-performance evidence, but quality still depends on question design and accessible administration.

Online timed examination

Can evidence: Time-bounded application or interpretation where an approved platform is appropriate.

Design for visibility: Use platform controls only where approved; vary data or prompts; require reasoning.

Caution: Do not describe lockdown or invigilation as eliminating AI access. Test accessibility, privacy, reliability and contingency arrangements.

Open-book timed assessment

Can evidence: Selection, synthesis and judgement under time constraints.

Design for visibility: Use supplied sources, comparative reasoning and explicit justification.

Caution: Open access increases uncertainty about assistance. Clear permissions and valid questions matter more than novelty.

Take-home essay or extended response

Can evidence: Argument, synthesis, disciplinary voice and evidence use.

Design for visibility: Use a specific intellectual problem, source justification, trade-offs and proportionate process evidence.

Caution: Do not rely on personalisation, unpublished material or a process log as proof. Align criteria with reasoning rather than polish.

Case study, policy memo or business proposal

Can evidence: Diagnosis, decision quality, practical judgement and audience awareness.

Design for visibility: Require a recommendation, assumptions, evidence, alternatives and trade-offs using supplied material.

Caution: Novel or local context can improve authenticity, but current models can analyse newly supplied information.

Problem set or quantitative task

Can evidence: Method selection, calculation, modelling assumptions and interpretation.

Design for visibility: Require workings, conceptual explanation, error checking and sensitivity or limitation analysis.

Caution: Parameter variation broadens assessment but does not prevent external assistance.

Coding or data-analysis project

Can evidence: Architecture, analytical judgement, testing, reproducibility and interpretation.

Design for visibility: Use version history, a demonstration or explanation where relevant; identify substantive AI-generated code.

Caution: A development history supports process evidence but is not automatically authentic. Protect data and clarify permitted coding assistance.

Group project

Can evidence: Collaboration, integrated product and individual contribution.

Design for visibility: Use milestones, roles, contribution statements and an individual evidence component when appropriate.

Caution: Avoid excessive monitoring. Clarify team and individual AI permissions and mark allocation.

Presentation or poster

- Can evidence:** Content selection, communication, visual design and oral response.
- Design for visibility:** Separate criteria; use proportionate Q&A or explanation; specify preparation and delivery permissions.
- Caution:** Live response can strengthen evidence of understanding but must be consistent, accessible and reliable.

Practical, clinical, laboratory or performed assessment

- Can evidence:** Real-time practical capability, judgement and professional communication.
- Design for visibility:** Observe the relevant performance using standardised rubrics; vary scenarios for breadth and security.
- Caution:** Direct observation is strong evidence of performance, but AI may still be part of preparation or professional tools and must be addressed explicitly.

Portfolio, fieldwork or placement evidence

- Can evidence:** Development over time, selection, reflection and context-specific learning.
- Design for visibility:** Use dated observations, artefacts, supervisor/client evidence and a justified curation rationale.
- Caution:** Protect confidentiality and avoid claiming that physical experience alone prevents AI-assisted write-up.

Dissertation, thesis or research project

- Can evidence:** Independent research judgement, methods, evidence, analysis and sustained development.
- Design for visibility:** Define permissions by stage; use supervision and milestones; require local declaration; apply research policy where relevant.
- Caution:** A viva can strengthen evidence but is not universally required or infallible. Consider accessibility, examiner consistency and programme rules.

— Language to use

AVOID	USE INSTEAD
Inherently AI-resistant	Provides strong direct evidence of [specific capability] under these conditions
AI cannot replicate this	This design makes [reasoning/performance/decision] more visible
Not in the training data	Requires engagement with supplied, specific evidence and disciplinary context
A viva proves authorship	Structured questioning can strengthen evidence of individual understanding
AI-blocking prevents access	Approved controls reduce opportunities but do not eliminate unauthorised assistance

— Implementation and approval checklist

- ✓ The written AI rule applies clearly to the discrete assessment and is issued in advance.
- ✓ The assessment purpose, permission level and marking criteria are aligned.
- ✓ Equality of baseline access is provided where a specific AI tool is authorised or required.
- ✓ Oral, invigilated, digital or process components are accessible, proportionate and locally approved.
- ✓ Students know what to declare and how the declaration will be used.
- ✓ No sensitive, identifiable, confidential or restricted material is placed in an inappropriate AI service.
- ✓ The assessment is reviewed using student experience, marking evidence, operational workload and current guidance.

Official sources

AI use in summative assessment: governance.admin.ox.ac.uk/education-committee/policies/ai-use-in-summative-assessment
Generative AI at Oxford: ox.ac.uk/gen-ai
Information security: infosec.ox.ac.uk/use-generative-ai-services-such-as-chatgpt-safely
Full interactive toolkit: gitcausuk.github.io/aitoolkit